

Application Description:

This app is a habit tracker that helps users to consistently do certain activities. It does so by keeping track of the habits a user is achieving and notifying them when they are not achieving such habits. The goal of the app is to give users a way of tracking their progress, therefore giving them a way of keeping themselves accountable and helping them stay consistent in the habits they want to accomplish.

The main screen of the apps consists of a widget for each different habit. Each separate habit has its own card in its own color, showing the name of the habit as well as a checkbox for whether it was completed today. The app also allows for a view of recent history and stats calculating how well users are sticking to their habits, allowing them to understand their progress.

Features:*Habits List*

The app lets the user maintain a list of habits they want to track. The list is scrollable so that the user can add as many habits as they need to. Each habit is displayed on the main screen with its own tracker card and its own color that the user can use. Each habit comes with its own edit option that allows for changing the name of the habit, as well as the color of the card for the habit, and the notification settings. The app also includes an “Add Habit” button as well as giving the option to delete habits.

Notifications

The app allows for users to set up a daily notification for specific habits in the habit tracker. This sends a reminder at a specified time to remind the user to do it instead of letting them forget.

History and Stats

The app has a history feature that shows the user’s previous habit data. Each habit card shows small boxes representing a certain window of previous days with different colors signifying completion of the habit or not. The user is able to move left and right through the windows to look at their previous completion data.

On the right side of the card as well is their statistics. This section includes their 7-day and 30-day statistics to let them know how they are doing overall with keeping up on completing their habits.

Export to Gmail

The app also includes an export to gmail feature that lets the user export their previous progress to gmail. When the user presses the export button the app opens gmail with a pre-filled message containing their habit progress summary and a day-by-day progression from the last 30 days.

Tips for Maintaining Habits Feature

The habit tracker also includes links to James Clear’s (author of *Atomic Habits*) own website and his tips and tricks for achieving success in maintaining habits. Allowing these resources gives users a better understanding of how to keep doing their habits in a sustainable manner.

Security Model:

The main security assets of the app that we need to be looked after are the users' habit data, the reminder settings/notification permission, usage of Webview, and the exportation of the progress summaries. In general the app follows the principle of least privilege, only requesting permissions that are directly needed for its features, specifically having to request users before giving notifications.

To start off, the habit data includes the names of their habits, completion status, the reminder settings, and the colors. While this is not as sensitive as something like medical or financial data, it is still user information and so it should be protected. To do this we store the data locally using Room in the app's private internal storage. This means that it's stored in application data, instead of the shared external storage. This protects the storage in the app because it doesn't allow for other apps or malicious actors to alter/view any of the user's data.

The WebView feature (used to view tips for creating habits) can also introduce security risks. To combat risks from using WebView, we restrict the WebView to only the intended domain, as well as disabling JavaScript, file access, content access, and opening any new links outside of the app. The app's export to Gmail feature is an example of inter-application communication. To reduce the risk of sending to any other applications the export is only allowed to send to Gmail and has to be triggered by direct user action.

The apps trust that Android's normal security protections are functioning correctly, including the app sandboxing, private internal storage, and the permission system. It also assumes that the device using the app isn't rooted. If the device is rooted, this could compromise the protection of the locally stored data. The app also trusts that Gmail is trusted and safe when the user decides to export their progress to Gmail, and that the James Clear website is a trusted resource as well.